EE4623: Design Exercise

EE4623: ***CMOS Process and Device Simulation***

- **Objective:** **For the 1 μm NMOS transistor, optimize* its device characteristics but maintain the same V_{T0} as that for the 2 μm device. For this purpose, perform splits in the initial template file (see below for details) in order to simulate multiple values of relevant fabrication process parameters, as well as devices with increasingly smaller gate lengths: 10, 5, 2, 1, 0.5, 0.2, 0.1 μm, respectively.**
  **NOTE: See page 3 for the definition of the "optimized" device.*

- **Important rules & points to consider:**
  - ***The Final Report must be submitted*** **by A TEAM of students (1 report per team).** **This is to optimize the usage of a limited nr. of available SWB licenses, minimize the (typically long) simulation time, and maximize the time for simulations by multiplexing the availability of the students in a team.**
  - **Hence, students must form groups/teams. *Each team should have about 4…5 students (min. 3 students, MAX. 6!).*** **E-mail me the NAMES of all students in each group/team, at the LATEST BY 21/22/23 Sept. 2026 for groups F61, F62, F63, respectively.**
  - **The Report must:**
    - **Be AT LEAST 50 pages long.**
    - **Comprise the results of your simulations.** **Any introduction and/or theoretical analysis and calculations are most welcome. Show your obtained results in Figures and/or Tables, as most suitable. At the same time, also try to explain their meaning, AND *WHY* they behave in the observed manner (especially when comparing results after varying a parameter, or different devices). The command codes of the simulations can be included in Appendices;**
    - **Provide clear descriptions of the contribution of each student in the team.**
    - **Be submitted ONLY in *soft copy*!**
    - **Be submitted together with a Turnitin receipt which shows the degree of similarity with other sources;**
    - **Be accompanied by mutual peer review assessments in NTULearn by each student for all the other students in the group (TBD);**
    - **Be submitted at THE LATEST by 16 Nov. 2026 for all groups (F61, F62, F63), respectively.**

- **General approach:**
  - **To achieve the stated target, use as initial template the `CMOS_130nm-v0` exercise. First copy it in your own folder and rename it. Use THIS file as your "master template" for usage, to subsequently copy & modify it & (re)save it (with different names) as many times as necessary, or as a template from which to copy & paste the desired instructions. *For instance, create a new Project of the Hierarchical type. Then add tools (SProcess & SDevice) in which you will copy & paste the desired parts of code from the initial* `CMOS_130nm-v0` *exercise. Recommended actions for your own subsequent simulations:***
    - ***In the Table Pane (Simulation Tree) delete the lines referring to PMOS.***
    - ***In the SProcess command file replace the multiple ion implantations used in each fabrication step with a SINGLE one.***
    - ***Similarly, modify at least the gate oxidation by using a single line of a code for a simple oxidation command instead of the existing multi-step oxidation***

with ramp-up & ramp-down before & after the actual oxidation.

- ***Make sure that you define at the beginning of the code all variables that you will indicate as parameters in the table (e.g., time & temperature of gate oxidation, dose & energy of implant) and use to make splits.***
- ***Also, make as comments the steps for realizing the LDD regions.***
- ***For better understanding about how the code can be modified, see/use my exercise files (`2D-LOCOS-v1`, `2DGS_Diode2v1` and `JFETlong2B`).***

❑ **Suggestions about HOW to proceed** [may be applied & carried out in your simulations]:

❖ Study the **(ideal) theoretical V_{TH} equation** and understand each term in its expression and understand its *functional dependence* on each physical parameter as well as the *approximations involved* that lead to the subsequent, more practical expression of V_T. Based on the gate work function and fixed oxide charge (specified in simulation) and the resulting structure (i.e., gate oxide thickness t_{ox} and channel doping profile):

- Estimate the "average" channel doping level and use this average doping to calculate the long-channel threshold voltage V_{T0}. Compare the calculated V_{T0} with that from the numerical device.
- What is the "effective" channel doping (N_{eff} or $\overline{N}$ instead of the N_A used in the V_T equation) that would give the same V_{T0} from the equation as that from the numerical device? Understand the meaning of the "effective" channel doping [and how to obtain it] and the approximations involved.

❑ With reference to the initial SWB example, implement the process to simulate only the NMOS transistor. Carry out a **design of experiment (DOE)** whose resulting structures will be used to investigate the device *scaling characteristics* and performance optimization with *process variations*. For this purpose:

- Use *split-wafer experiments* to vary the most relevant process parameters, e.g.:
  - DIfferent oxide thickness t_{ox} by varying oxidation time t and/or temperature T,
  - Change the V_T value by varying the surface (or peak) doping concentration N_S (N_{peak}) or $\overline{N}$ by varying the dose Q and/or energy E of the V_T-adjust implant step.

  Other parameters, for other process steps could also be considered (see below). These variations will enable you to deduce (and plot) the variations of the threshold voltage V_T versus t_{ox}, V_T versus Q and E, etc. **The initial "default" values for the 2 μm NMOS transistor are:**
  - **Substrate: *p*-type, doped with B, with resistivity of 10 Ω·cm (corresponds to a doping concentration $N_A \cong 1.35\times10^{15}$ cm^{-3});**
  - **Gate oxide thickness t_{ox} = 300 Å;**
  - **V_T-adjust implant dose and energy: $Q = 2\times10^{12}$ cm^{-2} and E = 30 keV, respectively;**
  - **Theoretical expected threshold voltage value $V_{T0} \cong$ 0.7…0.8 V (an average value of ~0.75 V can be considered).**
- Collect relevant data from device cross-sectional views at various process steps, probe doping profiles through different cutlines;

❑ Use the results of simulations for devices with increasingly smaller gate lengths to observe and optimize the device performance through **target–**

variable relationship. Use analytical theory as a guide to relate target–variable dependencies and do *graphical solutions* (numerical interpolation) for data analysis:

- Examine (with your own simulations) the *transistor I–V characteristics*: understand functional (or behavioural) dependence of I_{DS} on V_{GS} at both low and high V_{DS}, and study the dependence of V_{T0}, V_{TS}, g_m, and S_t on I_{ON}, I_{OFF}. Observe, analyze, and understand the basic behaviours ("shapes") of:
  - Linear (@V_{D0}) vs. saturation (@V_{DD}) I_{DS}–V_{GS} on linear scale;
  - Linear (@V_{D0}) vs. saturation (@V_{DD}) I_{DS}–V_{GS} on log scale;
  - Long-channel vs. short-channel I_{DS}–V_{GS} on linear/log scales;
  - Long-channel vs. short-channel I_{DS}–V_{DS} curves for various V_{GS} values, on linear scale.
- Examine the *transistor scaling characteristics*, i.e. the variation of device parameters (V_{T0}, V_{TS}, I_{ON}, I_{OFF}, I_{CRIT}, g_{mS}, S_{t0} and S_{tS}) as a function of gate length L, namely for its various decreasing values (remember that the actual "metallurgical" effective device channel length is $L_{eff} = L - 2\sigma x_j$, where $\sigma \cong 0.7$ is due to n^+ source/drain lateral diffusion).
  - Write a simple program (Matlab or C, or just export data into Excel) to plot V_T vs. L curves based on the theoretical V_T equation (long-channel) and "triangle" charge-sharing model (short-channel) with a given set of the *transistor's process parameters* (N_A, t_{ox}, Φ_M; x_j). Vary and observe the V_T vs. L changes by changing parameter values.
  - Obtain *technology optimization curves*, e.g., V_{T0}–log(L), V_{TS}–log(L), g_m–log(L), S_t–log(L), I_{on}–log(L), log(I_{OFF})–log(L), and log(I_{off})–I_{ON}. Compare with theoretical calculations, e.g., V_{T0} "roll-off" based on the simple "triangle" *charge-sharing model*. Determine minimum gate length based on a given set of criteria; or alternatively, determine various performance parameters at a given "designed" gate length.

❖ Determine *new* process variables for improved "on/off" current trade-off for logic-circuit applications. Specifically, aim that the **designed L = 1 µm device is "optimized", i.e. its drive current (I_{ON}) would be increased AND leakage current (I_{OFF}) can be reduced, with the guide that the linear threshold voltage V_{T0} at L = 1 µm is *relatively* unchanged** (i.e. remains at the theoretical expected value of ~0.7…0.8 V) **and has minimal "roll-off" & DIBL**. Other parameters (e.g., S_{t0} and S_{tS}, g_{mMax} and g_{mS}, g_{dsS}, R_{on}, etc.) may also be considered IF so desired (but may make the optimization more difficult) Below are some examples of DOE's you may follow.

- For the long-channel (10 µm) and short-channel (1 µm) devices, run separate DOEs for both *V_T-adjustment implant dose* (Q_{impl}) variations at a given gate oxidation *time* (t), and for t variations at a given Q_{impl}, respectively. Plot the *"scattered" optimization data* such as log(I_{OFF}) vs. I_{ON}, S_{tS} vs. g_{mS}, V_{TS} vs. V_{T0}. Observe the variation trend of "V_{T0} roll-off" (defined as $\Delta V_{T0} = V_{T0_10\mu m} - V_{T0_1\mu m}$) and "drain-induced barrier lowering (DIBL)" ($\Delta V_{DIBL} = V_{T0_1\mu m} - V_{TS_1\mu m}$) as a function of *time* t (or t_{OX}) as well as a function of Q_{impl} (or N_A). [*Hint*: Do some calculations as a guide]. For the short-channel (1 µm) device, ultimately identify/develop the best

process recipe (Q_{impl} and t and other parameters, as you think necessary), where "best" refers to the "optimized" definition explained in the previous blue paragraph above. Understand & explain the reason why the found/developed process is the best, based on the S_{tS} vs. g_{mS} and V_{TS} vs. V_{T0} variations and other such data.

- Based on the ideas in reducing the two major short-channel effects, namely, "ΔV_{T0} roll-off" and "ΔV_{DIBL}", decrease t_{OX} (by decreasing gate oxidation *time* t) and increase N_A (by increasing V_T-adjustment implant *Dose*, Q_{impl}), while maintaining the same V_{T0} for the L = 1 μm device.
- Based on the best-estimated combination of values for the most important parameters dictating the transistor behaviour (i.e. Q_{impl} and t), run the "full-loop" for various gate-length devices in comparison with the original process. Plot *technology curves* for the "new" process with respect to the "original" process, such as **V_{T0} and V_{TS} vs. log(L)**; g_{mS} vs. log(L); S_{tS} vs. log(L); I_{ON} vs. log(L); log(I_{OFF}) vs. log(L); and **log(I_{OFF}) vs. I_{ON}**.
- *Optional*: consider other process variables, such as punch-through (PT) implant dose/energy; and/or n^+ source/drain implant dose/energy, and/or add the steps for the realization of the LDD regions.

- ❑ As was indicated previously, collect relevant data to explain why your new process is better than the original given process. Relate the process variations to the reduction of major short-channel effects.
- ❑ Once certain improvements are made, find out the reasons and follow similar approaches to see if you can <u>further</u> optimize the process to improve the performance (i.e., take your new process as "nominal"); therefore, to understand how the I_{ON} and I_{OFF} are dependent on the device and process parameters and how they can be both (further?) improved.

- NOTE: The basic MOS transistor performance parameters given at the theory taught in this 2nd part of EE4623 are: linear and saturation threshold voltages (V_{T0} and V_{TS}), on-state saturation current (I_{ON} or I_{DSsat}), off-state leakage current (I_{OFF}), subthreshold swing (S_t), transconductance (g_m), and drain (output) conductance (g_{ds}). Their definitions are:

  Linear threshold voltage: $V_{T0} \equiv V_{GS} \mid I_{DS} = 0 \text{ @ } g_{m0} = \max\ (V_{DS} = 0.05\text{V})$
  Critical current: $I_{CRIT} = I_{DS} \text{ @ } V_{GS} = V_{T0}\ (V_{DS} = 0.05\text{ V})$
  Saturation threshold voltage: $V_{TS} \equiv V_{GS} \mid I_{DS} = I_{CRIT}\ (V_{DS} = 5\text{ V})$
  On-state saturation current: $I_{ON} \equiv I_{DSsat} \mid V_{GS} = V_{DS} = 5\text{ V}$
  Off-state leakage current: $I_{OFF} \equiv I_{DS} \mid V_{GS} = 0,\ V_{DS} = 5\text{ V}$
  Subthreshold swing (linear): $S_{t0} \equiv \Delta V_{GS}/\Delta\log(I_{DS})=1\text{ dec [mV/dec]}\ (V_{DS} = 0.05\text{V})$
  Subthreshold swing(saturation): $S_{tS} \equiv \Delta V_{GS}/\Delta\log(I_{DS})=1\text{ dec [mV/dec]}\ (V_{DS} = 5\text{V})$
  Saturation transconductance: $g_{mS} \equiv \mathrm{d}I_{DS}/\mathrm{d}V_{GS} \text{ @ } V_{GS} = V_{DS} = 5\text{ V}$
  Saturation drain (output) conductance: $g_{dsS} \equiv \mathrm{d}I_{DS}/\mathrm{d}V_{DS} \text{ @ } V_{GS} = V_{DS} = 5\text{ V}$

All the above are at zero body bias: $V_{BS} = 0$. For body effect, the threshold voltage is defined as: $V_T(V_{BS}) \equiv V_{GS} \text{ @ } I_{DS} = I_{CRIT}\ (V_{DS} = 0.05\text{ V},\ V_{BS} = -5\text{ V})$.